



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal to NAPCMS

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

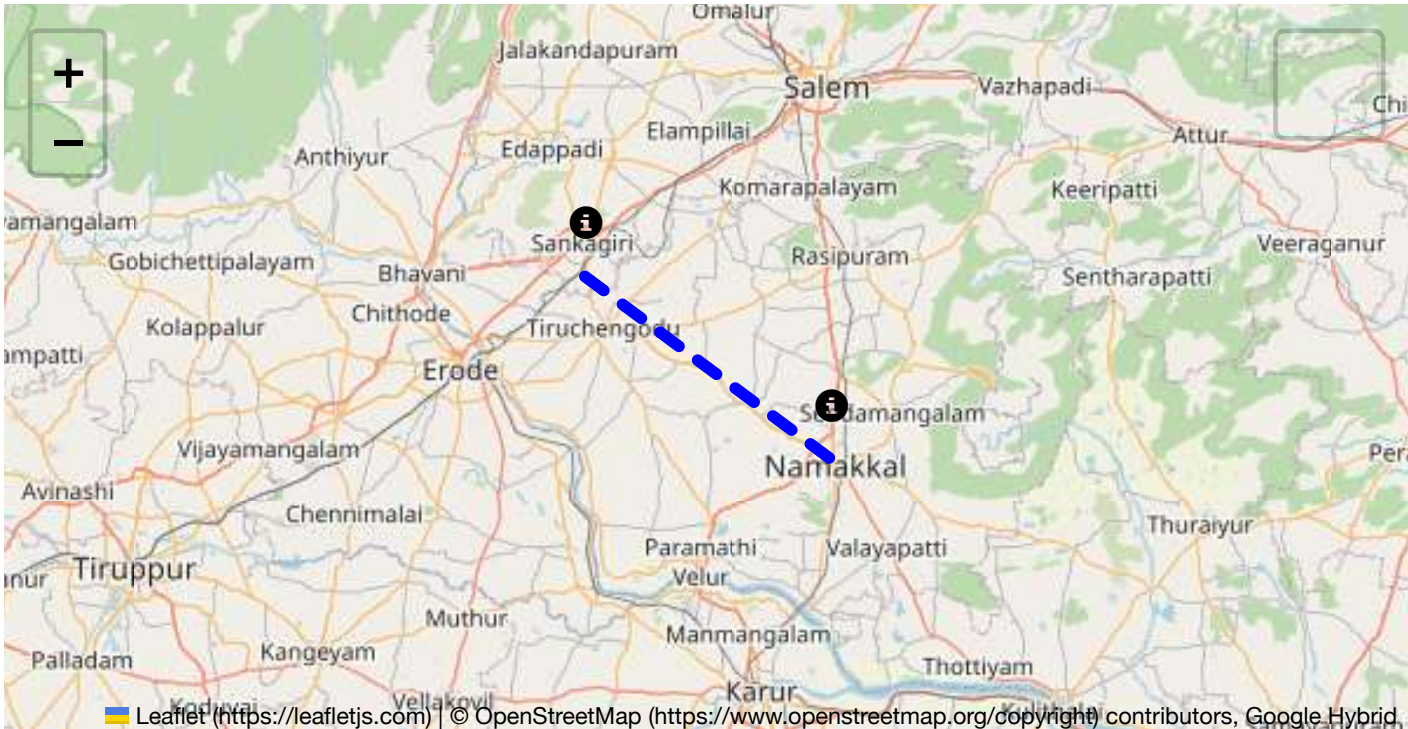
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

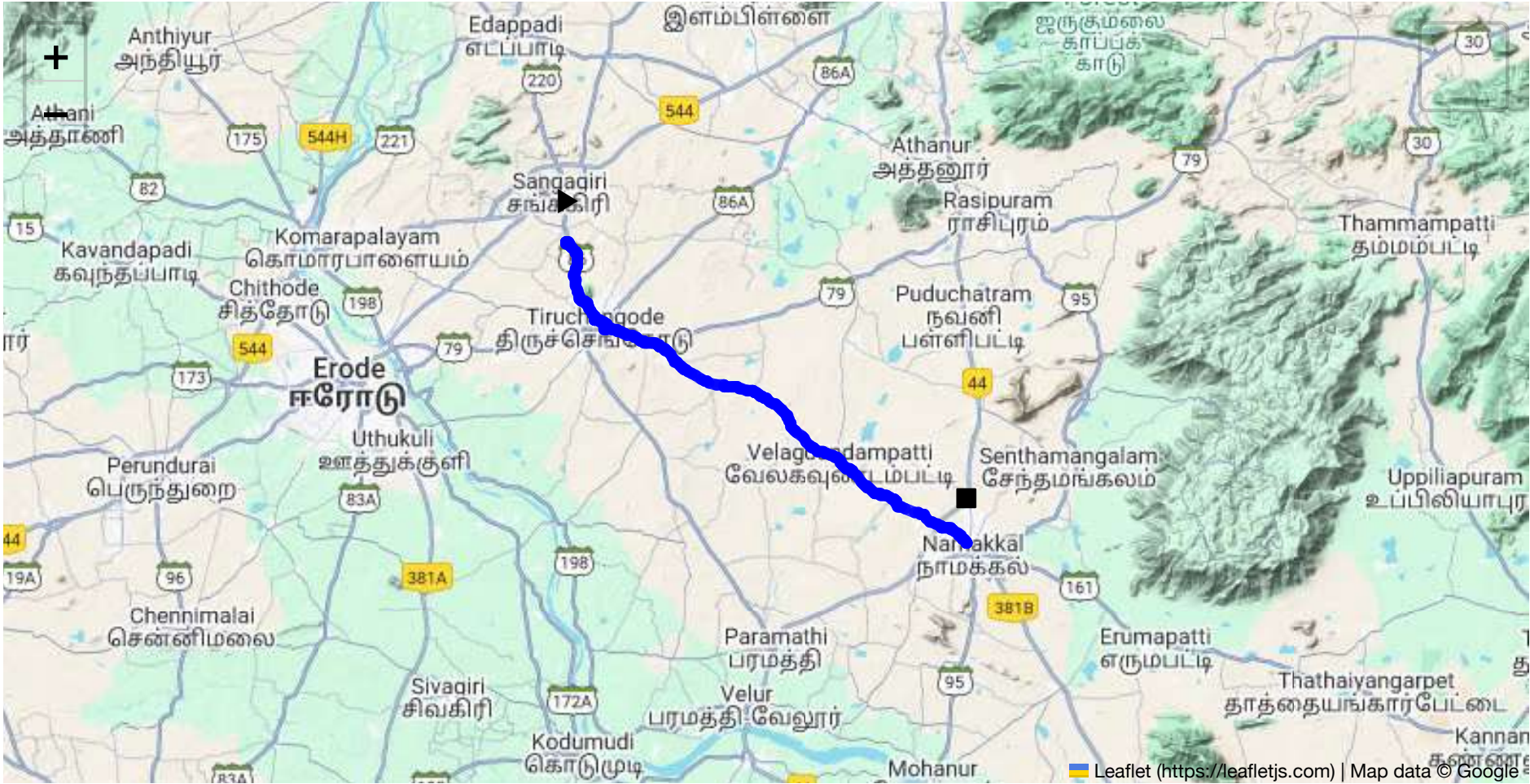
Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 43.38 km
Estimated Duration: 1.0 hours
Adjusted Duration (Heavy Vehicle): 1.2 hours

Start: (11.4381, 77.8734)
End: (11.22787, 78.161527)



Welcome to the Journey Risk Management Study

Route Safety Analysis for Transport from Sangagiri to Namakkal, Tamil Nadu

1. Overview of the Route Map

The route covers approximately 43.38 kilometers from Sangagiri to Namakkal. It primarily involves travel along rural and regional roads, with connection to State Highway 94 (SH 94). The drive is expected to take roughly one hour, depending on traffic and road conditions.

2. Typical Weather Conditions and Potential Weather-Related Hazards

The region experiences a tropical climate with hot summers, moderate rainfall during the monsoon, and mild winters.

- **Potential Hazards:** Heavy rainfall during monsoons (June to September) can lead to waterlogged roads and decreased visibility. Heat waves in the summer can also affect vehicle performance.

3. Analysis of Traffic Patterns

- **Peak Hours:** Common congestion is during morning hours (7:00 AM - 9:00 AM) and evening hours (5:00 PM - 7:00 PM), particularly near urban centers.
- **Congestion-Prone Areas:** Entrances into Namakkal and intersections with SH 94 can often be congested.

4. Assessment of Road Quality and Infrastructure

- **Road Quality:** Generally maintained but can be narrow in some rural areas. Potholes and uneven surfaces may be present, especially post-monsoon.
- **Infrastructure:** Limited shoulder space and few rest areas; watch for local traffic and pedestrians in rural stretches.

5. Suggestions for Alternative Routes

- **In Emergencies:** SR 94 serves as the primary alternative route; however, local secondary roads can provide diversions in cases of major blockages.

6. Summary of Local Regulations Affecting Hazardous Material Transport

- **Local Regulations:** Mandated use of specific transport permits for hazardous materials. Restrictions likely apply during school commute hours near urban areas.

7. Overview of Historical Incidents

- **Historical Incidents:** Documentation of past incidents is limited, but typical issues include vehicle breakdowns and minor road accidents. Few major incidents have involved heavy vehicles or hazardous materials.

8. Environmental Considerations and Sensitive Areas

- **Sensitive Areas:** Agricultural zones and nearby villages; spills could severely affect local ecosystems and communities.
- **Environmental Impact:** Strict adherence to environmental guidelines is required to prevent contamination.

9. Communication Coverage

- **Communication Dead Zones:** Reliable coverage through most of the route, but some rural segments might experience weak signals. It is advisable to have secondary communication devices.

10. Estimated Emergency Response Times

- **Response Time Estimates:** Rural areas may experience delays, with response times stretching from 30 minutes to an hour. Urban areas, such as Namakkal, have quicker emergency services.

11. Overall Summary of Risk Assessment

The route is relatively straightforward but demands vigilance due to potential weather disruptions, road quality issues, and urban congestion. There is a moderate risk associated with the transport of hazardous materials due to environmental sensitivity and communication gaps. Adequate preparation and adherence to local regulations are crucial to ensure safety.

By covering these aspects, the truck driver can be well-informed and prepared for the journey, ensuring maximum safety and compliance throughout the transport operation.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	11.38395, 77.89480	15 KM/Hr
1	U-Turn	High	11.2269168, 78.1625554	10 KM/Hr
2	Turn	High	11.43811, 77.87348	15 KM/Hr
3	Turn	Medium	11.43956, 77.87340	30 KM/Hr
4	Turn	High	11.43968, 77.87345	15 KM/Hr
5	Blind Spot	Blind Spot	11.44029, 77.87544	10 KM/Hr
6	Turn	Medium	11.38342, 77.89975	30 KM/Hr
7	Turn	Medium	11.38190, 77.90254	30 KM/Hr
8	Turn	High	11.38186, 77.90258	15 KM/Hr
9	Blind Spot	Blind Spot	11.37841, 77.90140	10 KM/Hr
10	Blind Spot	Blind Spot	11.22692, 78.16256	10 KM/Hr
11	Turn	High	11.22688, 78.16250	15 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
0	hospital	Tiruchengode, Goverment Hospital	11.3903328, 77.8920627	30 km/h	Medium
1	hospital	SPM Medical Centre,Tiruchengode	11.3881331, 77.8931963	30 km/h	Medium
5	hospital	Jayaa Hospital	11.3848223, 77.9021886	30 km/h	Medium
8	hospital	Vivekananda Medical Care Hospital	11.364562, 77.9482668	30 km/h	Medium
10	hospital	Manickampalayam, Government Hospital	11.32887, 78.017713	30 km/h	Medium
15	hospital	Rudhra Hospital	11.226569, 78.163216	30 km/h	Medium
16	hospital	Aruthra Hospital	11.2264666, 78.1631524	30 km/h	Medium
17	clinic	Sri Ram Nursing Home, Namakkal	11.2259986, 78.1631944	30 km/h	Medium
18	clinic	Thangavel Skin Care	11.2253765, 78.1644957	30 km/h	Medium
19	hospital	Rajendran Children's Hospital	11.2251887, 78.1649142	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
20	clinic	Lung Care Centre	11.2276789, 78.1638586	30 km/h	Medium
21	hospital	Vignesh Hospital	11.224818, 78.1621315	30 km/h	Medium
22	hospital	Venkat Hospital	11.2253645, 78.1646159	30 km/h	Medium
23	hospital	Shree Meenakshi Dental Hospital	11.225269, 78.164275	30 km/h	Medium
26	clinic	Vishnu Ayurveda Clinic	11.224262, 78.164173	30 km/h	Medium
27	clinic	Dhanvanthiri Physiotheraphy	11.2245256, 78.1638528	30 km/h	Medium
28	clinic	Chandra Test Tube Baby Centre	11.2241806, 78.1645931	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
2	school	அரசு ஆண்கள் மேல்நிலைப் பள்ளி	11.3850439, 77.8948279	30 km/h	Medium
3	school	அரசு பெண்கள் மேல்நிலைப் பள்ளி	11.3844247, 77.8949053	30 km/h	Medium
4	marketplace	திருச்செங்கோடு தினசரி காய்கறி சந்தை	11.3833608, 77.8970145	30 km/h	Medium
6	college	Vivekanandha Educational Institutions for Women	11.3596633, 77.9435556	30 km/h	Medium
7	college	Vivekanandha Dental College	11.3643705, 77.9475897	30 km/h	Medium
9	college	Vivekanandha College of engineering for Women	11.3565938, 77.9465342	30 km/h	Medium
11	college	The Surabi College of Polytechnic	11.2660726, 78.0939552	30 km/h	Medium
12	college	CMS College Of Engineering	11.2647508, 78.0969044	30 km/h	Medium
13	school	Porasapalayam Government School	11.2504918, 78.1163024	30 km/h	Medium
14	school	Sri Vinayaga Higher Secondary School	11.2479837, 78.1182908	30 km/h	Medium
24	school	Government Elementary School	11.224555, 78.163967	30 km/h	Medium
25	school	Government Secondary School	11.2244426, 78.1647533	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.38395, 77.89480



Risk Type: U-Turn

Risk Level: High

Speed Limit: 10 KM/Hr

Coordinates: 11.2269168, 78.1625554



Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr
Coordinates: 11.44029, 77.87544



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.38342, 77.89975



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.38190, 77.90254



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.38186, 77.90258



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.37841, 77.90140



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.22692, 78.16256



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.22688, 78.16250
